

CoreFusion Technologies

Master End-to-End System Workflow & Architecture Blueprint

Document Target: Complete 27-Phase Specification, Portal Workflows, API Map & Data Models
Tech Stack: React + Vite (Frontend), FastAPI + SQLAlchemy (Backend), PostgreSQL + Supabase Auth
System Scope: 4 System Portals, 9 Employee Workspaces, 42 Routers, 45 Database Models

1. High-Level Architecture & Tech Stack

CoreFusion operates on a multi-portal client-server architecture. The frontend is built with React 18 and Vite, communicating over a REST API with a high-performance Python FastAPI backend backed by PostgreSQL. Authentication is federated via Supabase Auth with server-side JWKS validation and local RBAC enforcement.

Layer	Technology / Framework	Key Specifications & Configurations
Frontend UI	React 18, Vite, Tailwind CSS	33 routes, 4 portals, 10 role tabs, SessionStorage auth storage
Backend API	FastAPI, Pydantic, Async SQLAlchemy	42 routers, 45 ORM models, uniform envelope {success, data, message}
Database & Auth	PostgreSQL (Supabase), Alembic	UUID primary keys, 5 migration revisions, JWKS verification, PgBouncer pool
Cloud Infra	Vercel (FE), Render (BE)	Frontend: cf-azure-eta.vercel.app BE: Render HTTP service

2. Portals, Role Hierarchy & Access Matrix

Role Hierarchy: super_admin > admin > hr, project_manager > Employee Roles (sales, marketing, developer, qa, support, finance) > client

Portal	Route Path	Authorized Roles	Primary Portal Capabilities
Public Site	/	Public / Unauthenticated	CMS content, services, industries, blog, contact form → lead creation
Client Portal	/client	client	Projects, invoices, payments, support tickets, shared files, meetings
Employee Portal	/employee, /sales, /dev...	All employee roles, PM, HR, Admin	10 base tabs (Attendance, Timesheets, Leaves) + role-specific workspaces
Admin Panel	/admin	admin, super_admin	CMS CRUD, User/Employee/Client management, Reports, Audit logs
Super Admin	/super-admin	super_admin	Tenant management, global system metrics, security logs & permissions

- Employee Role Workspaces:**
- **Sales:** CRM Dashboard, Leads, Proposals, Contracts, Clients, Meetings, Reports
 - **Project Manager:** Team Projects, Kanban Task Board, Milestone & Delivery Approvals

- **Developer / QA:** Assigned Development Tasks, Sprint Tracking, Test Queue
- **Support:** Support Ticket Queue, SLA Monitor, Client Reply Dispatch
- **Finance:** Invoice Generator, Payment Recording, Auto-Reconciliation
- **HR:** Leave Approvals, Recruitment, Timesheet Verification, Employee Documents

3. API Map & Database Architecture (45 Models)

The system relies on 45 SQLAlchemy models across 8 distinct business domains. All API endpoints reside under /api/v1 with uniform error and payload handling via CRUDBase[Model].

Domain	Core Tables / Models	Key Fields & Schema Attributes
Core Auth & RBAC	users, roles, permissions, role_permissions	id (UUID), email, role, is_active, slug, module, action
HR & Employee	employees, attendance, leaves, timesheets, payslips, employee_documents	employee_code, department_id, check_in/out, hours, gross/net_pay, rating
CRM & Sales	leads, proposals, contracts, contact_submissions	company, contact_name, estimated_value, proposal_version, client_signed_at
Operations & PM	clients, projects, tasks, project_members	client_id, account_manager_id, progress_percent, priority, due_date
Finance	invoices, payments	invoice_number (INV-ts), amount, tax, total_amount, payment_method, status
Support & Ops	tickets, ticket_replies, meetings, notifications, audit_logs	ticket_number (TCK-ts), priority, sla_due_at, meeting_link, action_type
CMS & Content	blogs, services, case_studies, testimonials, careers, events, etc.	title, slug, content, is_published, meta_title, author_id, view_count

4. 27-Phase Complete End-to-End Business Workflows

Phase 1: Public Website → Contact Form → Lead Capture

Workflow: Visitor submits contact form on website (name, email, phone, company, department, service_id, industry_id, budget, requirements). Backend rate-limits (5/min), saves ContactSubmission, and automatically provisions a Lead with source='website', mapping budget to estimated_value and requirements to notes.

Key Endpoints: POST /api/v1/contact | **Models:** contact_submissions, leads

Phase 2: Lead Evaluation & Qualification

Workflow: Sales reps review new leads in Sales Workspace. Pipeline stages: new → contacted → requirement_gathering → proposal_sent → proposal_approved → converted | disqualified. Sales reps capture meeting notes, confirm requirements, timeline, evaluation result, or rejection reason directly on the Lead record.

Key Endpoints: GET/PATCH /api/v1/leads/{id} | **Models:** leads

Phase 3: Lead-to-Client Account Conversion

Workflow: Upon agreement, Sales converts Lead to Client. Backend executes atomic operation: creates User account (role='client'), creates Client profile linking account_manager_id, and marks Lead status as 'converted'.

Key Endpoints: POST /api/v1/leads/{id}/convert | **Models:** leads, users, clients

Phase 4: Credential Email & Client Onboarding

Workflow: System sends welcome email to newly created Client containing portal login URL, temporary password/setup token, and assigned Account Manager contact details.

Key Endpoints: POST /api/v1/auth/provision-client | **Models:** users, notifications

Phase 5: Client Portal & Data Isolation

Workflow: Client logs in at /client. All GET requests automatically filter by client_id == current_user.client.id to ensure absolute strict multi-tenant isolation. Client views dashboard, active projects, invoices, files, and tickets.

Key Endpoints: GET /api/v1/clients/me/* | **Models:** clients, projects, invoices

Phase 6: PM & Employee Project Setup

Workflow: PM receives converted project request, creates Project record (title, budget, deadline), assigns team members via project_members, breaks scope into Tasks with assigned employees and due dates.

Key Endpoints: POST /api/v1/projects, POST /api/v1/tasks | **Models:** projects, tasks, project_members

Phase 7: Proposal Workflow & Review Gate

Workflow: Sales drafts proposal → submits for PM review → PM approves scope/cost → sent to Client. Client views proposal in portal → signs digitally. Dual-signature generates binding Contract.

Key Endpoints: POST /api/v1/proposals, POST /api/v1/contracts/{id}/sign | **Models:** proposals, contracts

Phase 8: Project Tracker & Daily Updates

Workflow: Developers log daily progress updates against assigned tasks. Progress percentage auto-calculates based on task completion and feeds directly into internal PM dashboard.

Key Endpoints: PATCH /api/v1/tasks/{id}/status | **Models:** tasks, projects

Phase 9: Project Completion & Delivery Approval

Workflow: PM marks project 100% complete and uploads deliverable package. Client receives delivery alert in portal, reviews deliverables, and clicks 'Approve Delivery' or 'Request Changes'.

Key Endpoints: PUT /api/v1/projects/{id}/deliver | **Models:** projects, client_files

Phase 10: Invoicing, Payments & Auto-Reconciliation

Workflow: Finance creates invoice (auto-generated number INV-{ts}). Total computed as amount + tax. Client pays via portal (Stripe/Bank). When cumulative completed payments $\geq$ total_amount, invoice auto-status flips to 'paid'.

Key Endpoints: POST /api/v1/finance/invoices, POST /payments | **Models:** invoices, payments

Phase 11: Document Management & Secure Downloads

Workflow: Clients and employees upload documents. Files stored in secure storage. Download endpoints verify JWT ownership before streaming file bytes.

Key Endpoints: POST /api/v1/media/upload, GET /media/download | **Models:** media, client_files

Phase 12: Support Ticket Lifecycle & SLA Tracking

Workflow: Client submits ticket (auto-numbered TCK-{ts}). Ticket enters Support Queue. System calculates SLA response due time. Support staff posts replies and marks ticket 'resolved'.

Key Endpoints: POST /api/v1/tickets, POST /tickets/{id}/replies | **Models:** tickets, ticket_replies

Phase 13: Meeting Scheduling & Video Links

Workflow: Sales/PM schedules meeting with Client/Lead. System records date, time, agenda, and video URL, broadcasting calendar invites to participants.

Key Endpoints: POST /api/v1/meetings | **Models:** meetings

Phase 14: Employee Portal & Role Workspaces

Workflow: Employees access /employee. Unified dashboard renders 10 base HR tabs (attendance, leaves, timesheets, payslips) + specialized workspace tabs based on authenticated role.

Key Endpoints: GET /api/v1/employees/me/profile | **Models:** employees, users

Phase 15: Attendance, Clock-In & Leave Approvals

Workflow: Employee clocks in/out daily. Requests leave (sick/casual). HR reviews pending leave queue and approves/rejects with comments, auto-updating leave balances.

Key Endpoints: POST /employees/me/attendance, POST /leaves | **Models:** attendance, leaves

Phase 16: Dual Project Views (Internal vs External)

Workflow: System maintains two distinct views for every project: Internal view (detailed developer tasks, internal notes, hour logs) vs Client view (milestone summaries, public updates, invoices).

Key Endpoints: GET /projects/{id} vs GET /clients/me/projects | **Models:** projects, tasks

Phase 17: Timesheets & Time Tracking

Workflow: Developers log hours worked per project daily. PM/HR reviews weekly timesheet submissions and approves hours for billing and payroll calculation.

Key Endpoints: POST /employees/me/timesheets | **Models:** timesheets

Phase 18: Performance Reviews, Goals & Training

Workflow: HR creates periodic performance reviews with ratings and goal metrics. Employees browse internal course catalog and enroll in training programs.

Key Endpoints: GET /employees/me/performance-reviews | **Models:** performance_reviews, courses

Phase 19: Payroll & Payslip Generation

Workflow: HR generates monthly payslips based on base salary, approved leaves, and tax deductions. Employees securely view and download PDF payslips.

Key Endpoints: GET /employees/me/payslips | **Models:** payslips

Phase 20: Employee Onboarding & Compliance Documents

Workflow: Onboarding employees upload tax forms, IDs, and contracts. HR verifies documents and flags compliance status.

Key Endpoints: POST /employees/me/documents | **Models:** employee_documents

Phase 21: Admin Panel & Global Platform Management

Workflow: Admins access /admin to monitor platform KPIs, manage user accounts, assign roles, inspect audit logs, and trigger system reports.

Key Endpoints: GET /api/v1/admin/*, GET /audit-logs | **Models:** users, audit_logs, reports

Phase 22: RBAC & Granular Permission Enforcement

Workflow: System verifies role permissions on every request via FastAPI require_roles() dependency and optional fine-grained action checks.

Key Endpoints: FastAPI Dependency Gate | **Models:** roles, permissions, role_permissions

Phase 23: CMS Content Lifecycle

Workflow: Marketing/Admin drafts blog posts, services, case studies, and testimonials. Content moves from draft to published, triggering site update.

Key Endpoints: POST/PUT /api/v1/blogs, /services, /case-studies | **Models:** blogs, services, case_studies

Phase 24: Services Centralization

Workflow: Admin manages centralized service catalog. Services link dynamically to contact form dropdowns, pricing packages, and portfolio items.

Key Endpoints: GET/POST /api/v1/services | **Models:** services

Phase 25: Industries Vertical Management

Workflow: Marketing creates industry solution pages (Healthcare, Fintech, E-commerce). Associated case studies automatically filter by industry ID.

Key Endpoints: GET/POST /api/v1/industries | **Models:** industries, case_studies

Phase 26: Blog Management & Comment Moderation

Workflow: Authors write articles. Readers submit comments on public blog posts. Admin moderates comments (approve/spam/delete) before publication.

Key Endpoints: POST /api/v1/blogs/{id}/comments | **Models:** blogs, comments

Phase 27: System Audit Logs & Activity Monitoring

Workflow: Every sensitive mutation (user edit, financial transaction, role change) automatically writes an entry to audit_logs recording actor ID, action, timestamp, and IP.

Key Endpoints: Middleware / Router hooks | **Models:** audit_logs

5. End-to-End Execution Sequence Diagrams

5.1 CRM Pipeline (Lead → Proposal → Contract → Client Account)

Step	Actor	Action & API Operation	State Transition
1	Visitor / Sales	Submit Contact Form or Create Lead POST /api/v1/contact or POST /leads	Lead created (Status: new)
2	Sales Rep	Draft & Send Proposal to Client POST /proposals → PUT /proposals/{id}	Proposal status: draft → sent
3	Client	Review Proposal & Sign Contract POST /contracts/{id}/sign (client_signed=true)	Contract status: client_signed
4	System / Admin	Company Counter-signs & Provision Client POST /contracts/{id}/sign (provision_client=true)	Client Account Active Lead status: converted

5.2 Financial Invoicing & Payment Auto-Reconciliation

Step	Actor	Action & API Operation	System Calculation & Status
1	Finance / Admin	Generate Client Invoice POST /api/v1/finance/invoices	Invoice auto-numbered INV-{ts} Total = Amount + Tax (Status: draft)
2	Finance	Dispatch Invoice to Client PATCH /finance/invoices/{id}	Invoice status: sent Visible in Client Portal
3	Client / System	Submit Payment Installment POST /finance/invoices/{id}/payments	Payment recorded Cumulative sum evaluated
4	System Auto-Gate	Reconcile Total Payments vs Invoice Amount	If Cumulative ≥ Total: Status auto-flips to paid

6. Implementation Audit Scorecard (Backend vs Frontend)

Executive Audit Summary: The backend implementation is highly mature (92% complete across all 27 spec areas). The primary focus area is connecting completed backend endpoints to corresponding frontend UI views.

§	Workflow Area	Backend Status	Frontend Status	Implementation Notes
1	Public Site & Contact Lead	Complete	Partial	Form needs service/industry/budget fields added to match API
2	Lead Evaluation	Complete	Partial	Evaluation result fields present in DB; need UI form wiring
3-4	Lead Conversion & Auth	Complete	Partial	Backend handles atomic creation; welcome email integration ready
5	Client Portal Isolation	Complete	Partial	Ownership scopes active on all endpoints; portal UI operational
6-7	Project Setup & Proposals	Complete	Partial	Dual signing flow built; PM review gate to be exposed in UI

8-9	Project Tracker & Delivery	Complete	Missing	Backend fully built (updates, milestones); needs client approval UI
10	Payments & Invoicing	Complete	Partial	Auto-payment reconciliation complete; PDF invoice download ready
12	Support Tickets & SLA	Complete	Partial	Ticket creation & replies functional; SLA counter active in API
14-17	Employee Portal & HR	Complete	Complete	Attendance, leave approvals, timesheets fully integrated
21-22	Admin & RBAC Control	Complete	Complete	Role hierarchy enforced across routers; frontend guards active
23-27	CMS & System Audit	Complete	Complete	Services, blogs, case studies, audit logging fully operational

7. Operations & Environment Quick Reference

Frontend Dev Setup: cd frontend && npm install && npm run dev (Port 5173)
Backend Dev Setup: cd backend && uvicorn main:app --reload (Port 8000)
Database Migrations: alembic upgrade head
Environment Keys: VITE_API_URL, VITE_SUPABASE_URL, DATABASE_URL, SUPABASE_JWT_SECRET

Approved by: CoreFusion Engineering Team & Systems Architecture Group
This document serves as the authoritative complete workflow definition for the CoreFusion platform.